STEEL BEAM

Beam analysis.

Analyze a simply supported steel beam under uniform load.

> **For preliminary evaluation only.** This analysis is not a construction approval or a complete member design. Connections, bearing, stability and scaffold-specific requirements require separate review.

ARRANGEMENT

SCHEMATIC

Beam arrangement

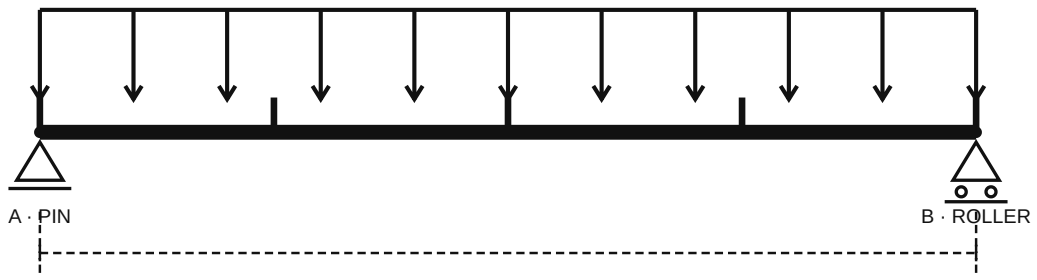
MODEL / 01

Beam geometry & loading

SCHEMATIC · NOT TO SCALE

ELEVATION · FULL-SPAN UNIFORM LOAD

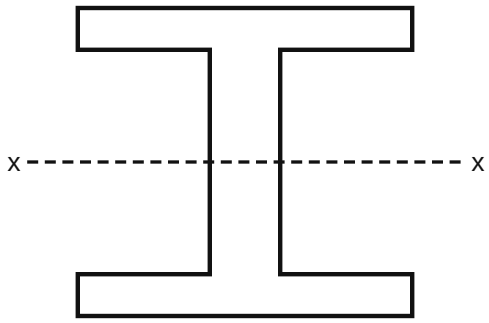
D 0.2 + L 0.8 kip/ft · self-weight included



L = 20 ft · DISTANCE BETWEEN SUPPORTS

BRACES / X 0, 5, 10, 15, 20 ft from A

SECTION · MAJOR-AXIS BENDING



STRONG AXIS · X–X

SECTION **W16X31**	07 / ELASTIC MODULUS E **29000 ksi**
08 / STRONG-AXIS I_x **375 in⁴**	09 / STIFFNESS FACTOR **0.8 × EI**

Analysis results

Responses for the load case shown above.

Service load case: 1.0D + 1.0L — results and diagrams below.

Support reaction · each	Maximum moment	Nominal EI deflection	Reduced EI deflection
10.31 kip	**51.55** kip-ft	**0.3413** in	**0.4266** in

ASD — no additional factor

AISC 360-16 ASD · A992 F_y = 50 ksi · nominal E = 29,000 ksi · service 1.0D + 1.0L · C_b = 1.0

Flexure · major axis

Meets listed check

Flange compactness	$b_f / 2t_f \le 0.38\sqrt{(E / F_y)}$	**6.284 ≤ 9.152** AISC B4.1b
Web compactness	$(d - 2k) / t_w \le 3.76\sqrt{(E / F_y)}$	**51.695 ≤ 90.553** AISC B4.1b
Plastic moment	$F_y Z_x / 12 = 50 \times 54 / 12$	**225 kip-ft** AISC F2-1
L_p	$1.76 r_y \sqrt{(E / F_y)} / 12$	**4.133 ft** AISC F2-5
L_r	$1.95 r_{ts} E/(0.7F_y) \times \sqrt{[J/(S_x h_o) + \sqrt{((J/(S_x h_o))^2 + 6.76(0.7F_y/E)^2)}]} / 12$	**11.833 ft** AISC F2-6
Governing segment	5–10 ft · inelastic LTB	**L_b = 5 ft**
C_b	Specified	**1.000**
Nominal moment	$\min[M_p, C_b(M_p - (M_p - 0.7F_y S_x / 12)(L_b - L_p)/(L_r - L_p))]$	**215.164 kip-ft** AISC F2-2
Allowable moment	M_n / Ω_b = 215.164 / 1.67	**128.841 kip-ft** AISC F2
Moment ratio	M / M_{allow} = 51.55 / 128.841	**0.400**

Unbraced segments · service load (moments in kip-ft)

Segment (ft)	L_b (ft)	C_b	M	M_n	M_{allow}	Ratio
0–5	5	1.000	38.663	215.164	128.841	0.300
5–10	5	1.000	51.55	215.164	128.841	0.400
10–15	5	1.000	51.55	215.164	128.841	0.400
15–20	5	1.000	38.663	215.164	128.841	0.300

Shear · major axis

Meets listed check

Web slenderness	$h / t_w \le 2.24\sqrt{(E / F_y)}$	**51.695 ≤ 53.946** AISC G2.1(a)
Web area	$A_w = d \times t_w = 15.9 \times 0.275$	**4.373 in²** AISC G2.1
Nominal shear	$0.6 F_y A_w C_{v1} = 0.6 \times 50 \times 4.373 \times 1$	**131.175 kip** AISC G2-1
Allowable shear	V_n / Ω_v = 131.175 / 1.5	**87.45 kip** AISC G2.1(a)
Shear ratio	V / V_{allow} = 10.31 / 87.45	**0.118**

ASD — 4:1 comparison

Illustrative only. Not a verified OSHA or ASD load combination; maximum intended load and failure-capacity basis require review.

Bending load	D + self-weight + (SF / 1.67)L = 0.2 + 0.031 + (4 / 1.67 × 0.8)	**2.1472 kip/ft**
Moment demand	max[w x(L − x) / 2], segment 5–10 ft	**107.358 kip-ft**
Moment ratio	107.358 / 128.841 kip-ft	**0.833** AISC F2
Shear load	D + self-weight + (SF / 1.50)L = 0.2 + 0.031 + (4 / 1.50 × 0.8)	**2.3643 kip/ft**
Shear demand	wL / 2 = 2.3643 × 20 / 2	**23.643 kip**
Shear ratio	23.643 / 87.45 kip	**0.270** AISC G2.1(a)

Not evaluated: bearing, local web forces, connections, axial force, torsion, deflection limits, brace design, and scaffold-specific requirements. No overall adequacy is determined; verify section data independently.

Service analysis

Section	W16X31 · E = 29,000 ksi	**I_x = 375 in⁴**
Beam self-weight	Section weight / 1,000 = 31 / 1,000	**0.031 kip/ft**
Uniform load	D + self-weight + L = 0.2 + 0.031 + 0.8	**1.031 kip/ft**
Reaction at each support	wL / 2 = 1.031 × 20 / 2	**10.31 kip**
Maximum moment	$wL^2 / 8 = 1.031 \times 20^2 / 8$	**51.55 kip-ft**
Nominal EI deflection	$5wL^4 / (384EI) = 5 \times (1.031 / 12) \times 240^4 / (384 \times 29{,}000 \times 375)$	**0.3413 in**
Reduced EI deflection	$\delta_{nominal}$ / 0.8	**0.4266 in**
Lateral braces	Measured from left support; not vertical supports	**0, 5, 10, 15, 20 ft**

Beam bending only; shear deformation excluded. No overall adequacy or independent verification. Bearing, connections, and scaffold-specific capacity are not evaluated. AISC v15 section data is from a third-party mirror; verify against the official source.

Project: Untitled project · Prepared by: Not specified · 2026-11-10T04:01:29.671Z · Engine 0.1.0

ⓘ **Independent software comparison not completed**

Checks are limited to the stated steel limit states; scaffold requirements require separate evaluation.

Response diagrams

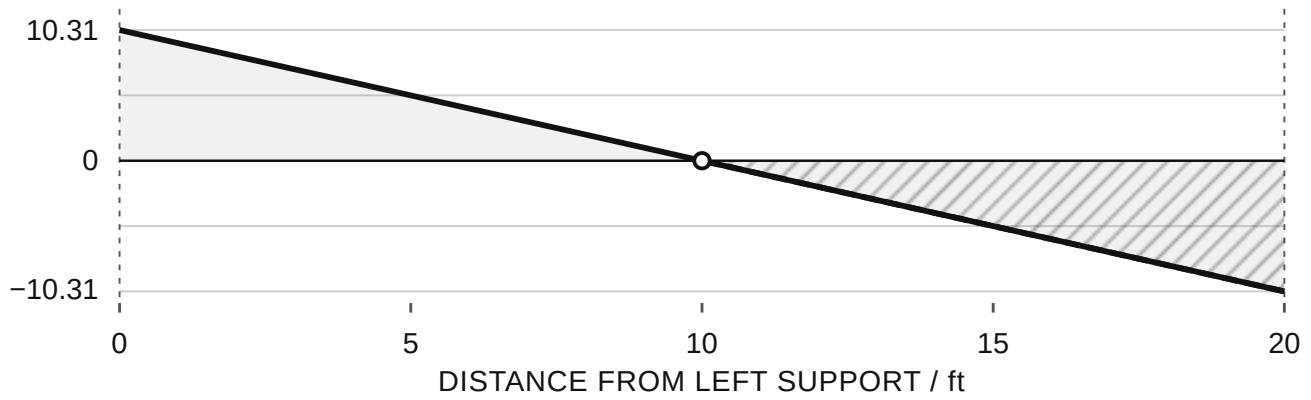
View how forces and displacement change along the span.

Service loads only. Positive shear equals the left reaction minus downward loads to the left of the cut; positive moment is sagging. Deflection is downward and uses reduced EI. Negative values use dashed lines and hatching. Curves connect sampled values from the response equations.

SERVICE CASE / 1.0D + 1.0L

Shear force

V / kip



MAXIMUM	MINIMUM
10.31 kip at 0 ft	**-10.31 kip at 20 ft**

Zero shear at 10 ft.

SERVICE CASE / 1.0D + 1.0L

Bending moment

M / kip-ft

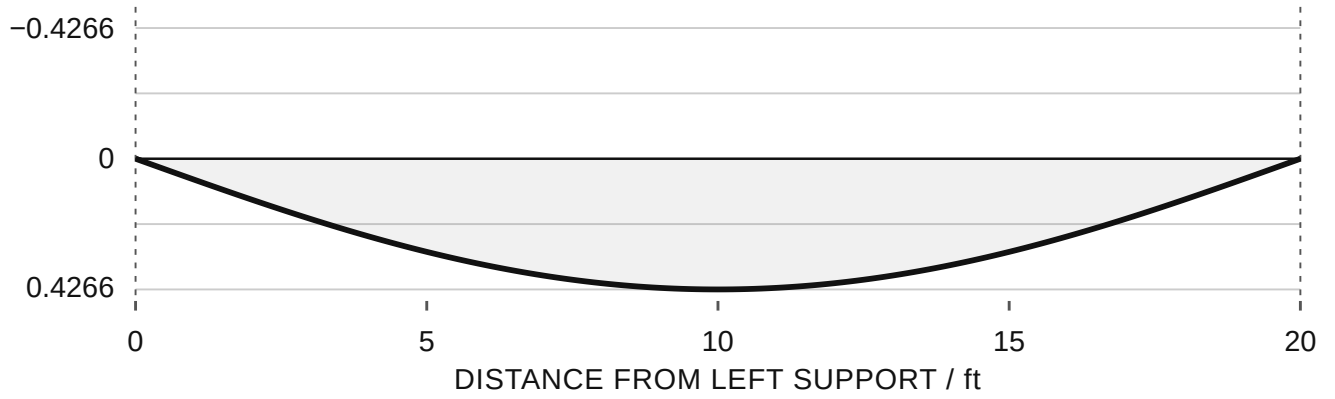


MAXIMUM	MINIMUM
51.55 kip-ft at 10 ft	**0 kip-ft at both supports**

SERVICE CASE / 1.0D + 1.0L

Bending deflection

δ / in



MAXIMUM	MINIMUM
0.4266 in at 10 ft	**0 in at both supports**

SECTION DATA

AISC v15 · mirrored

Section properties

AISC v15 properties from a third-party mirror; verify against the official source before use.

W **31** lb/ft	A **9.13** in²	d **15.9** in
bf **5.53** in	tw **0.275** in	tf **0.44** in
Ix **375** in⁴	Sx **47.2** in³	Zx **54** in³
Iy **12.4** in⁴	Sy **4.49** in³	Zy **7.03** in³
J **0.461** in⁴	Cw **739** in⁶	rx **6.41** in
ry **1.17** in	rts **1.42** in	ho **15.5** in

Terms of use · draft

This tool supports preliminary planning and does not replace review by a qualified professional. Verify inputs, applicable standards, connections, bearing, stability, and all scaffold-specific requirements before use. No certification or construction approval is provided.